

# MediTrack AI

## Medicine Reminder & Tracker Project Synopsis

### 1. Introduction

MediTrack AI is a Flutter-based healthcare management application designed to help users organize their medicines, receive timely reminders, monitor medication history, and improve medicine adherence. The application combines modern mobile development practices with Firebase services, local notifications, and API integration to provide a smooth healthcare experience.

### 2. Problem Statement

Many individuals forget medicine schedules due to busy lifestyles, multiple prescriptions, and irregular routines. The absence of proper tracking may lead to missed doses and poor medication management. MediTrack AI provides a digital solution to simplify medicine tracking and reminder management.

### 3. Objectives

- Develop a responsive and user-friendly Flutter mobile application.
- Provide automated medicine reminders and notification support.
- Maintain medicine records and dose history.
- Implement Firebase authentication and cloud data management.
- Create a professional healthcare-focused user experience.

### 4. Proposed Solution

The proposed system allows users to create personal medicine schedules, receive notifications at specified times, mark medicines as completed, and analyze their medication performance through history and statistics.

### 5. Major Modules

| Module         | Description                                                       |
|----------------|-------------------------------------------------------------------|
| Authentication | Secure user registration and login using Firebase Authentication. |

| Module              | Description                                                   |
|---------------------|---------------------------------------------------------------|
| Dashboard           | Displays today's medicines, progress, and upcoming reminders. |
| Medicine Management | Add, update, delete, and manage medicines.                    |
| Reminder System     | Schedules notifications for medicine timings.                 |
| History Tracking    | Maintains taken, missed, and skipped dose records.            |
| Profile Management  | Handles user details and application settings.                |

## 6. Technology Stack

Frontend: Flutter & Dart

Database: Cloud Firestore

Authentication: Firebase Authentication

State Management: Riverpod

Navigation: GoRouter

Local Storage: Shared Preferences

API Integration: REST API

Notifications: Flutter Local Notifications

## 7. UI/UX Design Approach

The application follows Material 3 design principles with responsive layouts, modern cards, smooth animations, light and dark themes, and an intuitive navigation system to provide a premium mobile experience.

## 8. Database Design

Firebase Firestore stores user profiles, medicine information, reminder schedules, and dose history using a structured cloud database architecture.

## 9. Future Enhancements

- AI-powered medicine assistance.
- Prescription scanning using OCR.
- Family member medicine management.
- Cloud push notifications.
- Doctor appointment integration.

## 10. Expected Outcome

The final application will act as a digital medicine assistant that improves medication discipline, reduces missed doses, and demonstrates practical implementation of Flutter, Firebase, API integration, and mobile application development.

## **11. Conclusion**

MediTrack AI represents a real-world healthcare solution developed using modern Flutter technologies. The project demonstrates complete mobile application development including UI design, database integration, notifications, and scalable architecture.